

Quadratic Forms

1.1 Definitions

Definition 1.1. For $\mathbf{x} \in \mathbb{R}^n$, a **quadratic form** with matrix $\mathbf{A}_{n \times n}$ is the function:

$$Q(\mathbf{x}) = \mathbf{x}' \mathbf{A} \mathbf{x}.$$

Without loss of generality, $\mathbf{A}$ is assumed symmetric, since the quadratic form induced by a non-symmetric matrix $\tilde{\mathbf{A}}$ is equivalent to that induced by the symmetric matrix $\mathbf{A} = \frac{1}{2}(\tilde{\mathbf{A}} + \tilde{\mathbf{A}}')$. ■

Definition 1.2. A matrix $\mathbf{A}$ is **positive semi-definite** if for $\forall \mathbf{x} \in \mathbb{R}^n, \mathbf{x} \neq \mathbf{0}$ the quadratic form $Q(\mathbf{x}) = \mathbf{x}' \mathbf{A} \mathbf{x} \geq 0$. If inequality is strict, then $\mathbf{A}$ is *positive definite*. ■

1.2 Properties

Remark 1.1. Recall from the spectral theorem that a symmetric matrix $\mathbf{A}_{n \times n}$ is uniquely expressible as $\mathbf{A} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}'$, where $\mathbf{U}$ is an *orthogonal eigenbasis* and $\mathbf{\Lambda}$ is a diagonal matrix of *eigenvalues*. Since:

$$\mathbf{u}'_i \mathbf{u}_j = \begin{cases} 1, & i = j, \\ 0, & i \neq j, \end{cases}$$

$\mathbf{A}$ may be viewed as a linear combination of projection operators:

$$\mathbf{A} = \sum_{i=1}^n \lambda_i \mathbf{u}_i \mathbf{u}'_i = \sum_{i=1}^n \lambda_i \mathbf{u}_i (\mathbf{u}'_i \mathbf{u}_i)^{-1} \mathbf{u}'_i = \sum_{i=1}^n \lambda_i \mathbf{P}_i,$$

where $\mathbf{P}_i = \mathbf{u}_i (\mathbf{u}'_i \mathbf{u}_i)^{-1} \mathbf{u}'_i$ is projection onto the image of the i th eigenvector $\mathbf{u}_i$. ♦

Proposition 1.1. For $\mathbf{x} \in \mathbb{R}^n$, the quadratic form $Q(\mathbf{x}) = \mathbf{x}' \mathbf{A} \mathbf{x}$ is expressible as:

$$Q(\mathbf{x}) = \sum_{i=1}^n \lambda_i (\mathbf{u}'_i \mathbf{x})^2,$$

where (λ_i) are the eigenvalues and $(\mathbf{u}_i)$ the eigenvectors of $\mathbf{A}$. ♦

Proof. Let $\mathbf{A}_{n \times n} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}'$. The quadratic form is expressible as:

$$Q(\mathbf{x}) = \mathbf{x}' \mathbf{A} \mathbf{x} = \text{tr}(\mathbf{x}' \mathbf{A} \mathbf{x}) = \text{tr}(\mathbf{x}' \mathbf{U} \mathbf{\Lambda} \mathbf{U}' \mathbf{x})$$

Let $\mathbf{y} = \mathbf{U}' \mathbf{x}$, then:

$$Q(\mathbf{x}) = \text{tr}(\mathbf{y}' \mathbf{\Lambda} \mathbf{y}) = \text{tr}(\mathbf{\Lambda} \mathbf{y} \mathbf{y}') = \sum_{i=1}^n \lambda_i y_i^2 = \sum_{i=1}^n \lambda_i (\mathbf{u}'_i \mathbf{x})^2$$

■

Proposition 1.2 (Rayleigh Quotient). For a symmetric matrix $\mathbf{A}_{n \times n}$:

$$\sup_{\{\mathbf{u}: \|\mathbf{u}\|^2=1\}} \mathbf{u}' \mathbf{A} \mathbf{u} = \max_{i=1}^n (\lambda_i),$$

where (λ_i) are the eigenvalues of $\mathbf{A}$. ◆

Proof. Let $Q(\mathbf{x}) = \mathbf{x}' \mathbf{A} \mathbf{x}$ denote a quadratic form induced by $\mathbf{A}$. Take the spectral decomposition $\mathbf{A} = \mathbf{U} \mathbf{\Lambda} \mathbf{U}'$. Since $\mathbf{U}$ is an eigenbasis for $\mathbb{R}^n$, every $\mathbf{x}$ is expressible as $\mathbf{x} = \mathbf{U} \boldsymbol{\beta}_x$, for some coefficient $\boldsymbol{\beta}_x$. Now:

$$Q(\mathbf{x}) = \sum_{i=1}^n \lambda_i (\mathbf{u}_i' \mathbf{x})^2 = \sum_{i=1}^n \lambda_i (\mathbf{u}_i' \mathbf{U} \boldsymbol{\beta}_x)^2 = \sum_{i=1}^n \lambda_i (\mathbf{e}_i' \boldsymbol{\beta}_x)^2 = \sum_{i=1}^n \lambda_i \beta_{x,i}^2 \leq \max_{i=1}^n (\lambda_i) \boldsymbol{\beta}_x' \boldsymbol{\beta}_x$$

The squared norm of the coefficient $\|\boldsymbol{\beta}_x\|^2$ is equal to the squared norm of the original vector $\mathbf{x}$:

$$\|\boldsymbol{\beta}_x\|^2 = \boldsymbol{\beta}_x' \boldsymbol{\beta}_x = \boldsymbol{\beta}_x' \mathbf{U}' \mathbf{U} \boldsymbol{\beta}_x = (\mathbf{U} \boldsymbol{\beta}_x)' \mathbf{U} \boldsymbol{\beta}_x = \mathbf{x}' \mathbf{x} = \|\mathbf{x}\|^2.$$

Rearranging the bound on the quadratic form gives:

$$\frac{\mathbf{x}' \mathbf{A} \mathbf{x}}{\boldsymbol{\beta}_x' \boldsymbol{\beta}_x} = \frac{\mathbf{x}' \mathbf{A} \mathbf{x}}{\mathbf{x}' \mathbf{x}} \leq \max_{i=1}^n (\lambda_i)$$

Defining $\mathbf{u} = \mathbf{x}/\|\mathbf{x}\|$ such that $\|\mathbf{u}\|^2 = \mathbf{u}' \mathbf{u} = 1$ gives $\mathbf{u}' \mathbf{A} \mathbf{u} \leq \max(\lambda_i)$. Finally, since the bound is independent of $\mathbf{x}$, conclude that:

$$\sup_{\{\mathbf{u}: \|\mathbf{u}\|^2=1\}} \mathbf{u}' \mathbf{A} \mathbf{u} = \max_{i=1}^n (\lambda_i).$$
■

The χ_ν^2 Distribution

2.1 Definitions

2.1.1 Central

Definition 2.1. If (Z_i) are independent $N(0, 1)$ random variates, then the sum:

$$\sum_{i=1}^{\nu} Z_i^2 \stackrel{d}{=} \chi_\nu^2(0)$$

follows a central χ^2 distribution with ν degrees of freedom. ■

Proposition 2.1. The distribution of a $\chi_1^2(0)$ random variable is gamma with shape parameter $\alpha = 1/2$ and rate parameter $\lambda = 1/2$. ◆

Proof. Let $Z \sim N(0, 1)$ and set $Y = Z^2$. The distribution of Y is:

$$P(Y \leq y) = P(Z^2 \leq y) = P(-\sqrt{y} \leq Z \leq \sqrt{y}) = \Phi(\sqrt{y}) - \Phi(-\sqrt{y}).$$

Differentiating to obtain the density:

$$f(y) = \phi(\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} + \phi(-\sqrt{y}) \cdot \frac{1}{2\sqrt{y}} = \phi(\sqrt{y}) \cdot \frac{1}{\sqrt{y}} = \frac{1}{\sqrt{2\pi}} y^{1/2-1} e^{-y/2}.$$

■

Proposition 2.2. The characteristic function of the $\chi^2_\nu(0)$ distribution is:

$$\phi(\omega) = (1 - 2i\omega)^{-\nu/2}.$$

◆

Proof. Suppose Z_i are IID $N(0, 1)$. The characteristic function of $Y = \sum_{i=1}^\nu Z_i^2$ is:

$$\phi_\nu(\omega) = \mathbb{E} \left(e^{i\omega \sum_{i=1}^\nu Z_i^2} \right) = \prod_{i=1}^\nu \mathbb{E} \left(e^{i\omega Z_i^2} \right) = \prod_{i=1}^\nu \phi_1(\omega),$$

where $\phi_1(\omega)$ is the characteristic function of the $\chi^2_1(0)$ distribution. Finding $\phi_1(\omega)$:

$$\begin{aligned} \phi_1(\omega) &= \mathbb{E} \left(e^{i\omega Z_i^2} \right) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{i\omega z^2} e^{-z^2/2} dz \\ &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{z^2}{2}(1+2i\omega)} dz = (1 - 2i\omega)^{-1/2}. \end{aligned}$$

■

Corollary 2.1. The central χ^2 distribution with n degrees of freedom is a gamma distribution with shape parameter $\alpha = n/2$ and rate $\lambda = 1/2$.

$$\chi^2_n(0) \stackrel{d}{=} G(\alpha = n/2, \lambda = 1/2).$$

♣

2.1.2 Non-Central

Definition 2.2. If (X_i) are independent $N(\mu_i, \sigma^2)$ random variables, then the sum:

$$\frac{1}{\sigma^2} \sum_{i=1}^\nu X_i^2 \stackrel{d}{=} \chi^2_\nu(\delta)$$

follows a non-central χ^2 distribution with ν degrees of freedom and non-centrality parameter:

$$\delta = \frac{1}{\sigma^2} \sum_{i=1}^\nu \mu_i^2.$$

■

Proposition 2.3. For an integer $k \geq 0$:

$$2^{2k}(k!)\Gamma(k + 1/2) = (2k)!\sqrt{\pi}.$$

◆

Proof. Using the recursion $\Gamma(k + 1) = k\Gamma(k)$:

$$\begin{aligned}\Gamma(k + 1/2) &= (k - 1/2)(k - 3/2) \cdots (1/2)\Gamma(1/2) \\ &= (1/2)^k(2k - 1)(2k - 3) \cdots (1)\sqrt{\pi} = (1/2)^k(2k - 1)!!\sqrt{\pi}.\end{aligned}$$

Observe that $(2k - 1)!!$ contains the odd terms from $(2k)!$, which is expressible as:

$$\begin{aligned}(2k)! &= 2k \cdot (2k - 1)(2k - 2)(2k - 3)(2k - 4) \cdots (2)(1) \\ &= (2k - 1)(2k - 3) \cdots (1) \cdot 2(k)2(k - 1) \cdots 2(1) \\ &= (2k - 1)!! \cdot 2^k(k!).\end{aligned}$$

Substituting $(2^k k!)^{-1}(2k)!$ for $(2k - 1)!!$ in the expression for $\Gamma(k + 1/2)$ gives:

$$\Gamma(k + 1/2) = 2^{-k}(2^k k!)^{-1}(2k)!\sqrt{\pi}.$$

■

Proposition 2.4. The density of $\chi_1^2(\delta)$ is:

$$f(y) = e^{-\delta/2} \sum_{k=0}^{\infty} \frac{(\delta/2)^k}{k!} \cdot \frac{y^{k-1/2} e^{-y/2}}{2^{k+1/2}\Gamma(k + 1/2)}.$$

◆

Proof. Suppose $Z \sim N(0, 1)$, and let $Y = (Z + \mu)^2$, where $\mu = \sqrt{\delta}$. The distribution of Y is:

$$\begin{aligned}P(Y \leq y) &= P\{(Z + \mu)^2 \leq y\} = P(-\sqrt{y} \leq Z + \mu \leq \sqrt{y}) \\ &= P(-\sqrt{y} - \mu \leq Z \leq \sqrt{y} - \mu) = \Phi(\sqrt{y} - \mu) - \Phi(-\sqrt{y} - \mu).\end{aligned}$$

Differentiating to obtain the density:

$$\begin{aligned}f(y) &= \phi(\sqrt{y} - \mu) \cdot \frac{1}{2\sqrt{y}} + \phi(-\sqrt{y} - \mu) \cdot \frac{1}{2\sqrt{y}} \\ &= \frac{1}{2\sqrt{2\pi y}} \left(e^{-(\sqrt{y}-\mu)^2/2} + e^{-(\sqrt{y}+\mu)^2/2} \right) \\ &= \frac{e^{-(y+\mu^2)/2}}{2\sqrt{2\pi y}} (e^{\mu\sqrt{y}} + e^{-\mu\sqrt{y}}).\end{aligned}$$

Using the Taylor series representation:

$$\frac{1}{2} (e^{\mu\sqrt{y}} + e^{-\mu\sqrt{y}}) = \frac{1}{2} \sum_{k=0}^{\infty} \left(\frac{(\mu\sqrt{y})^k}{k!} + \frac{(-\mu\sqrt{y})^k}{k!} \right) = \frac{1}{2} \sum_{k=0}^{\infty} \frac{\sqrt{y}^k \mu^k}{k!} \{1 + (-1)^k\}.$$

Observe that the summand vanishes for odd k , therefore:

$$\frac{1}{2} \sum_{k=0}^{\infty} \frac{\sqrt{y}^k \mu^k}{k!} \{1 + (-1)^k\} = \sum_{l=0}^{\infty} \frac{\sqrt{y}^{2l} \mu^{2l}}{(2l)!} = \sum_{l=0}^{\infty} \frac{y^l \mu^{2l}}{(2l)!}.$$

Substituting the power series representation into the density:

$$f(y) = \frac{e^{-(y+\mu^2)/2}}{\sqrt{2\pi y}} \sum_{k=0}^{\infty} \frac{y^k \mu^{2k}}{(2k)!} = \frac{e^{-\mu^2/2}}{\sqrt{2}} \sum_{k=0}^{\infty} \frac{\mu^{2k}}{(2k)! \sqrt{\pi}} \cdot y^{k-1/2} e^{-y/2}.$$

Applying the identity from the previous proposition:

$$\begin{aligned} f(y) &= e^{-\mu^2/2} \sum_{k=0}^{\infty} \frac{\mu^{2k}}{2^{2k+1/2} (k!) \Gamma(k+1/2)} \cdot y^{k-1/2} e^{-y/2} \\ &= \sum_{k=0}^{\infty} e^{-\mu^2/2} \frac{(\mu^2/2)^k}{k!} \cdot \frac{1}{2^{k+1/2} \Gamma(k+1/2)} y^{k-1/2} e^{-y/2}. \end{aligned}$$

Observe that $e^{-\mu^2/2} (\mu^2/2)^k / (k!)$ is the density of the Poisson distribution, with mean $\mu^2/2$, and that:

$$\frac{1}{2^{k+1/2} \Gamma(k+1/2)} y^{k-1/2} e^{-y/2} = \frac{1}{2^{k+1/2} \Gamma(k+1/2)} y^{(2k+1)/2-1} e^{-y/2}$$

is the density of the $\chi^2_{2k+1}(0)$ distribution. The non-central $\chi^2_1(\delta)$ is therefore a Poisson mixture of central χ^2_{2k+1} distributions. ■

Proposition 2.5. The characteristic function of the $\chi^2_\nu(\delta)$ distribution is:

$$\phi(\omega) = e^{i\omega\delta/(1-2i\omega)} \cdot (1-2i\omega)^{-\nu/2}. \quad (2.1)$$

◆

Proof. Let (Z_i) denote independent $N(0, 1)$, set $\mu = \sqrt{\delta}$, and define:

$$Y = (Z_\nu + \mu)^2 + \sum_{i=1}^{\nu-1} Z_i^2,$$

such that Y has a $\chi^2_\nu(\delta)$ distribution. The characteristic function of Y is:

$$\phi_Y(\omega) = \mathbb{E} \left(e^{i\omega \{ (Z_\nu + \mu)^2 + \sum_{i=1}^{\nu-1} Z_i^2 \}} \right) = \mathbb{E} \left(e^{i\omega (Z_\nu + \mu)^2} \right) \cdot \prod_{i=1}^{\nu-1} \mathbb{E} \left(e^{i\omega Z_i^2} \right) = \phi_{1,\delta}(\omega) \phi_1^{\nu-1}(\omega),$$

where $\phi_1(\omega)$ is the characteristic function of the $\chi_1^2(0)$ distribution, and $\phi_{1,\delta}(\omega)$ is the characteristic function of the $\chi_1^2(\delta)$ distribution. Finding the latter:

$$\phi_{1,\delta}(\omega) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{i\omega(z+\mu)^2} e^{-z^2/2} dz = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-\frac{1}{2}g(z)} dz,$$

where:

$$\begin{aligned} g(z) &= z^2 - 2i\omega(z + \mu)^2 = z^2 - 2i\omega(z^2 + \mu^2 + 2z\mu) \\ &= -2i\omega\mu^2 + z^2(1 - 2i\omega) - 4i\omega z\mu \\ &= -2i\omega\mu^2 + (1 - 2i\omega) \left\{ z^2 - \frac{4i\omega z\mu}{1 - 2i\omega} \right\}. \end{aligned}$$

Let $\alpha = (1 - 2i\omega)^{-1}2i\omega\mu$, then:

$$\begin{aligned} g(z) &= -2i\omega\mu^2 + (1 - 2i\omega)\{z^2 - 2\alpha\} \\ &= -2i\omega\mu^2 + (1 - 2i\omega)\{(z - \alpha)^2 - \alpha^2\} \\ &= -2i\omega\mu^2 + \frac{4\omega^2\mu^2}{1 - 2i\omega} + (1 - 2i\omega)(z - \alpha)^2 \\ &= -2\omega\mu^2 \left\{ i - \frac{2\omega}{1 - 2i\omega} \right\} + (1 - 2i\omega)(z - \alpha)^2 \\ &= \frac{-2i\omega\mu^2}{1 - 2i\omega} + (1 - 2i\omega)(z - \alpha)^2. \end{aligned}$$

Thus, $\phi_{1,\delta}(\omega)$ resolves to:

$$\phi_{1,\delta}(\omega) = e^{i\omega\mu^2/(1-2i\omega)} \cdot \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-(1-2i\omega)(z-\alpha)^2/2} dz = e^{i\omega\mu^2/(1-2i\omega)} \cdot (1 - 2i\omega)^{-1/2}.$$

■

Corollary 2.2. The density of the $\chi_\nu^2(\delta)$ distribution is:

$$f(y) = e^{-\delta/2} \sum_{k=0}^{\infty} \frac{(\delta/2)^k}{k!} \cdot \frac{y^{(2k+\nu)/2-1} e^{-y/2}}{2^{k+\nu/2} \Gamma(k + \nu/2)}, \quad (2.2)$$

which is a Poisson ($\mu^2/2$) mixture of $\chi_{2k+\nu}^2(0)$ distributions. ♣

Remark 2.1. The corollary is verified by showing the characteristic function of the $\chi_\nu^2(\delta)$ density in (2.2) is the characteristic function in (2.1). ♦

2.2 Properties

Proposition 2.6. The mean and variance of the $\chi_\nu^2(\delta)$ distribution are:

$$\begin{aligned} \mathbb{E}\{\chi_\nu^2(\delta)\} &= \delta + \nu, \\ \mathbb{V}\{\chi_\nu^2(\delta)\} &= 4\delta + 2\nu. \end{aligned}$$

♦

Proof. From the characteristic function, the moment generating function is:

$$M(t) = e^{t\delta/(1-2t)} \cdot (1-2t)^{-\nu/2}.$$

The cumulant generating function is:

$$K(t) = \frac{t\delta}{1-2t} - \frac{\nu}{2} \ln(1-2t).$$

The first derivative of K is:

$$\dot{K}(t) = \delta(1-2t)^{-1} + 2t\delta(1-2t)^{-2} + \nu(1-2t)^{-1}.$$

Evaluating at $t = 0$ gives the expectation:

$$\mathbb{E}\{\chi_\nu^2(\delta)\} = \dot{K}(0) = \delta + \nu.$$

The second derivative of K is:

$$\ddot{K}(t) = 2\delta(1-2t)^{-2} + 2\delta(1-2t)^{-2} + 8t\delta(1-2t)^{-3} + 2\nu(1-2t)^{-2}.$$

Evaluating at $t = 0$ gives the variance:

$$\mathbb{V}\{\chi_\nu^2(\delta)\} = \ddot{K}(0) = 4\delta + 2\nu.$$

■

Quadratic Forms in Normal Random Variables

Proposition 3.1. Suppose $\mathbf{y}_{n \times 1} \sim N(\boldsymbol{\mu}, \sigma^2 \mathbf{I}_{n \times n})$ and $\mathbf{P}_{n \times n}$ is the projection onto a linear subspace $V \subseteq \mathbb{R}^n$ of dimension $m = \dim(V) = \text{tr}(\mathbf{P}) \leq n$. Then:

$$Q(\mathbf{y}) = \sigma^{-2} \cdot \mathbf{y}' \mathbf{P} \mathbf{y} \sim \chi_m^2(\delta), \quad (3.1)$$

with non-centrality parameter (NCP) $\delta = \sigma^{-2} \cdot \boldsymbol{\mu}' \mathbf{P} \boldsymbol{\mu}$. ◆

Proof. Let $\mathbf{U} \boldsymbol{\Lambda} \mathbf{U}'$ denote the spectral decomposition of $\mathbf{P}$, then the quadratic form is expressible as: $Q(\mathbf{y}) = \sigma^{-2} \cdot \mathbf{y}' \mathbf{U} \boldsymbol{\Lambda} \mathbf{U}' \mathbf{y}$. Define $\mathbf{x} = \mathbf{U}' \mathbf{y}$, then since $\mathbf{U}$ is orthogonal, $\mathbf{x} \sim N(\mathbf{U}' \boldsymbol{\mu}, \sigma^2 \mathbf{I})$, which shows the components of $\mathbf{x}$ are independent.

Recall that all eigenvalues of a projection matrix are either 1 or 0. Since $\mathbf{P}$ is projection onto a subspace of dimension m , the first m eigenvalues are 1, and the remaining $(n-m)$ are 0. Now the quadratic form is expressible as:

$$Q(\mathbf{y}) = \sigma^{-2} \cdot \mathbf{x}' \boldsymbol{\Lambda} \mathbf{x} = \frac{1}{\sigma^2} \sum_{i=1}^n \lambda_i X_i^2 = \frac{1}{\sigma^2} \sum_{i=1}^m X_i^2.$$

where $X_i \sim N(\mathbf{u}'_i \boldsymbol{\mu}, \sigma^2)$. Conclude that $Q(\mathbf{y})$ follows a non-central $\chi_m^2(\delta)$ distribution, with non-centrality parameter:

$$\delta = \frac{1}{\sigma^2} \sum_{i=1}^m (\mathbf{u}'_i \boldsymbol{\mu})^2 = \frac{1}{\sigma^2} \sum_{i=1}^n \lambda_i (\mathbf{u}'_i \boldsymbol{\mu})^2 = \sigma^{-2} \cdot \boldsymbol{\mu}' \mathbf{U} \boldsymbol{\Lambda} \mathbf{U}' \boldsymbol{\mu} = \sigma^{-2} \cdot \boldsymbol{\mu}' \mathbf{P} \boldsymbol{\mu}.$$

■

Theorem 3.1. Suppose $\mathbf{y}_{n \times 1} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, and $\mathbf{A}_{n \times n}$ is symmetric, positive definite. The distribution of $Q(\mathbf{y}) = \mathbf{y}' \mathbf{A} \mathbf{y}$ is a mixture of non-central χ_1^2 distributions:

$$Q(\mathbf{y}) = \mathbf{y}' \mathbf{A} \mathbf{y} \sim \sum_{i=1}^n \lambda_i \chi_1^2(\delta_i), \quad (3.2)$$

where $\delta_i = \nu_i^2$, $\nu_i = \mathbf{e}'_i \mathbf{U}' \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\mu}$, $\mathbf{U}$ is the eigenbasis of $\boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2}$, and (λ_i) are the corresponding eigenvalues. $\square$

Proof. Let $\mathbf{y} = \boldsymbol{\Sigma}^{1/2} \mathbf{z} + \boldsymbol{\mu}$ where $\mathbf{z} \sim N(\mathbf{0}, \mathbf{I})$. Express $Q(\mathbf{y})$ as:

$$\begin{aligned} Q(\mathbf{y}) &= (\boldsymbol{\Sigma}^{1/2} \mathbf{z} + \boldsymbol{\mu})' \mathbf{A} (\boldsymbol{\Sigma}^{1/2} \mathbf{z} + \boldsymbol{\mu}) \\ &= (\mathbf{z} + \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\mu})' \boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2} (\mathbf{z} + \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\mu}). \end{aligned}$$

Let $\mathbf{U} \boldsymbol{\Lambda} \mathbf{U}'$ denote the spectral decomposition of $\boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2}$. Re-expressing the quadratic form with use of the spectral decomposition of $\mathbf{U} \boldsymbol{\Lambda} \mathbf{U}'$:

$$\begin{aligned} Q(\mathbf{z}) &= (\mathbf{z} + \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\mu})' \mathbf{U} \boldsymbol{\Lambda} \mathbf{U}' (\mathbf{z} + \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\mu}) \\ &= (\mathbf{U}' \mathbf{z} + \mathbf{U}' \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\mu})' \boldsymbol{\Lambda} (\mathbf{U}' \mathbf{z} + \mathbf{U}' \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\mu}). \end{aligned}$$

Let $\boldsymbol{\nu} = \mathbf{U}' \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\mu}$ and define $\mathbf{x} = \mathbf{U}' \mathbf{z} + \boldsymbol{\nu}$, then:

$$\mathbf{x} \sim \mathbf{U}' \mathbf{z} + \boldsymbol{\nu} \stackrel{d}{=} N(\mathbf{0}, \mathbf{I}) + \boldsymbol{\nu} \stackrel{d}{=} N(\boldsymbol{\nu}, \mathbf{I}).$$

It follows that the quadratic form is a weighted sum of non-central χ_1^2 distributions:

$$Q(\mathbf{x}) = \mathbf{x}' \boldsymbol{\Lambda} \mathbf{x} = \sum_{i=1}^n \lambda_i X_i^2 = \sum_{i=1}^n \lambda_i \chi_1^2(\delta_i),$$

where $X_i \sim N(\nu_i, 1)$, $\nu_i = \mathbf{e}'_i \boldsymbol{\nu} = \mathbf{e}'_i \mathbf{U}' \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\mu}$, and $\delta_i = \nu_i^2$. $\blacksquare$

Proposition 3.2. If $\mathbf{A}$ has a known Cholesky decomposition $\mathbf{A} = \mathbf{L} \mathbf{L}'$, then the eigenbasis $\mathbf{U}$ and eigenvalues $\boldsymbol{\Lambda}$ appearing in (3.2) may be obtained from the spectral decomposition of $\mathbf{L}' \boldsymbol{\Sigma} \mathbf{L}$. $\blacklozenge$

Proof. Using the Cholesky decomposition:

$$Q(\mathbf{y}) = \mathbf{y}' \mathbf{L} \mathbf{L}' \mathbf{y} = (\mathbf{L}' \mathbf{y})' \mathbf{I} (\mathbf{L}' \mathbf{y}).$$

Let $\mathbf{x} = \mathbf{L}' \mathbf{y} \sim N(\mathbf{L}' \boldsymbol{\mu}, \mathbf{L}' \boldsymbol{\Sigma} \mathbf{L})$. Applying (3.2), $\mathbf{U}$ is the eigenbasis of:

$$(\mathbf{L}' \boldsymbol{\Sigma} \mathbf{L})^{1/2} \mathbf{I} (\mathbf{L}' \boldsymbol{\Sigma} \mathbf{L})^{1/2} = (\mathbf{L}' \boldsymbol{\Sigma} \mathbf{L}).$$

■

Proposition 3.3. The matrix $\boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2}$ appearing in (3.2) and the matrix $\mathbf{A} \boldsymbol{\Sigma}$ have the same eigenvalues. ♦

Proof. Recall that matrices $\mathbf{A}$ and $\mathbf{B}$ are *similar* if there exists an invertible *change of basis* $\mathbf{S}$ such that $\mathbf{B} = \mathbf{S}^{-1} \mathbf{A} \mathbf{S}$, and that similar matrices have the same eigenvalues. If $\boldsymbol{\Sigma}$ is positive definite, then $\boldsymbol{\Sigma}^{1/2}$ exists and is invertible; thus $\mathbf{A} \boldsymbol{\Sigma}$ and $\boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2}$ are similar matrices: $(\mathbf{A} \boldsymbol{\Sigma}) = \boldsymbol{\Sigma}^{-1/2} (\boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2}) \boldsymbol{\Sigma}^{1/2}$. ■

Example 3.1. Suppose $\boldsymbol{\Sigma}_{n \times n}$ is symmetric, positive semi-definite, then by the spectral decomposition $\boldsymbol{\Sigma} = \mathbf{U} \boldsymbol{\Lambda} \mathbf{U}'$, where $\mathbf{U}$ is orthogonal and $\lambda_i \geq 0$ for each $i \in \{1, \dots, n\}$. The square root of $\boldsymbol{\Sigma}$ is $\boldsymbol{\Sigma}^{1/2} = \mathbf{U} \boldsymbol{\Lambda}^{1/2} \mathbf{U}'$ since:

$$\boldsymbol{\Sigma}^{1/2} \boldsymbol{\Sigma}^{1/2} = (\mathbf{U} \boldsymbol{\Lambda}^{1/2} \mathbf{U}') (\mathbf{U} \boldsymbol{\Lambda}^{1/2} \mathbf{U}') = \mathbf{U} \boldsymbol{\Lambda} \mathbf{U}' = \boldsymbol{\Sigma}.$$

If $\boldsymbol{\Sigma}$ is positive definite ($\lambda_i > 0$ for $\forall i$), then the inverse of $\boldsymbol{\Sigma}$ is $\boldsymbol{\Sigma}^{-1} = \mathbf{U} \boldsymbol{\Lambda}^{-1} \mathbf{U}'$ since:

$$\boldsymbol{\Sigma} \boldsymbol{\Sigma}^{-1} = (\mathbf{U} \boldsymbol{\Lambda} \mathbf{U}') (\mathbf{U} \boldsymbol{\Lambda}^{-1} \mathbf{U}') = \mathbf{I} = (\mathbf{U} \boldsymbol{\Lambda}^{-1} \mathbf{U}') (\mathbf{U} \boldsymbol{\Lambda} \mathbf{U}') = \boldsymbol{\Sigma}^{-1} \boldsymbol{\Sigma}.$$

It follows that $\boldsymbol{\Sigma}^{-1/2} = \mathbf{U} \boldsymbol{\Lambda}^{-1/2} \mathbf{U}'$ is the square root of $\boldsymbol{\Sigma}^{-1}$. Moreover, expressing each matrix in terms of $\mathbf{U}$ and $\boldsymbol{\Lambda}$ gives the identities:

$$\boldsymbol{\Sigma}^{-1/2} \boldsymbol{\Sigma} \boldsymbol{\Sigma}^{-1/2} = \mathbf{I}, \quad \boldsymbol{\Sigma}^{1/2} \boldsymbol{\Sigma}^{-1} \boldsymbol{\Sigma}^{1/2} = \mathbf{I}.$$

♠

Example 3.2. Suppose in (3.2) that $\mathbf{y}_{n \times 1} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, and that $\mathbf{A} = \boldsymbol{\Sigma}^{-1}$. Then:

$$\boldsymbol{\Sigma}^{1/2} \mathbf{A} \boldsymbol{\Sigma}^{1/2} = \boldsymbol{\Sigma}^{1/2} \boldsymbol{\Sigma}^{-1} \boldsymbol{\Sigma}^{1/2} = \mathbf{I}.$$

Consequently, $\mathbf{U} = \mathbf{I}$ and $\lambda_i = 1$ for $\forall i$. Now $\nu_i = \mathbf{e}_i' \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\mu}$, and:

$$Q(\mathbf{y}) = \mathbf{y}' \boldsymbol{\Sigma}^{-1} \mathbf{y} \sim \sum_{i=1}^n \chi_1^2(\delta_i) \stackrel{d}{=} \chi_n^2(\delta),$$

$$\delta = \sum_{i=1}^n \delta_i = \sum_{i=1}^n (\mathbf{e}_i' \boldsymbol{\Sigma}^{-1/2} \boldsymbol{\mu})^2 = \boldsymbol{\mu}' \boldsymbol{\Sigma}^{-1} \boldsymbol{\mu}.$$

♠

Example 3.3. Suppose in (3.2) that $\Sigma = \sigma^2 \mathbf{I}$, such that $\mathbf{y}_{n \times 1} \sim N(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$. Moreover, let $\mathbf{x} = \sigma^{-1} \cdot \mathbf{y}$, such that $\mathbf{x}_{n \times 1} \sim N(\sigma^{-1} \boldsymbol{\mu}, \mathbf{I})$. Consider the distribution of:

$$Q(\mathbf{x}) = \mathbf{x}' \mathbf{A} \mathbf{x} = \frac{\mathbf{y}' \mathbf{A} \mathbf{y}}{\sigma^2}.$$

By direct application of the theorem:

$$Q(\mathbf{x}) \sim \sum_{i=1}^n \lambda_i \chi_1^2(\delta_i),$$

where $\delta_i = \nu_i^2$, $\nu_i = \mathbf{e}_i' \mathbf{U}' \sigma^{-1} \boldsymbol{\mu} = \sigma^{-1} \mathbf{u}_i' \boldsymbol{\mu}$, $\mathbf{U}$ is the eigenbasis of $\mathbf{A}$, and (λ_i) are the corresponding eigenvalues. If in addition $\mathbf{A}$ is a projection matrix of dimension m , then the first m eigenvalues of $\mathbf{A}$ are 1, and the remainder 0, such that:

$$Q(\mathbf{x}) \sim \sum_{i=1}^m \chi_1^2(\delta_i) \stackrel{d}{=} \chi_m^2(\delta),$$

$$\delta = \sum_{i=1}^m \delta_i = \frac{1}{\sigma^2} \sum_{i=1}^m (\mathbf{u}_i' \boldsymbol{\mu})^2 = \frac{1}{\sigma^2} \sum_{i=1}^n \lambda_i \boldsymbol{\mu}' \mathbf{u}_i \mathbf{u}_i' \boldsymbol{\mu} = \frac{1}{\sigma^2} \boldsymbol{\mu}' \mathbf{A} \boldsymbol{\mu}.$$

Finally, if $\boldsymbol{\mu} = \mathbf{0}$ and $\mathbf{A} = \mathbf{I}$, which is a projection matrix, then:

$$Q(\mathbf{x}) = \mathbf{x}' \mathbf{x} = \frac{\mathbf{y}' \mathbf{y}}{\sigma^2} \sim \sum_{i=1}^n \chi_1^2(0) \stackrel{d}{=} \chi_n^2(0).$$

♠

Remark 3.1. For numerical calculation of $P\{Q(\mathbf{x}) > t\}$, see the `CompQuadForm` package in R. Let $\mathbf{U}$ and $\boldsymbol{\Lambda}$ denote the eigenbasis and eigenvalues obtained from spectral decomposition of $\Sigma^{1/2} \mathbf{A} \Sigma^{1/2}$. The required inputs are the eigenvalues $\boldsymbol{\Lambda}$, and the non-centrality parameters $\boldsymbol{\delta} = \boldsymbol{\nu}^{\odot 2}$, where $\boldsymbol{\nu} = \mathbf{U}' \Sigma^{-1/2} \boldsymbol{\mu}$ and $\boldsymbol{\nu}^{\odot 2}$ denotes the element-wise square. If $\boldsymbol{\mu} = \mathbf{0}$, then $\boldsymbol{\delta} = \mathbf{0}$, and the eigenvalues are more easily obtained from $\mathbf{A} \Sigma$. ♦

Theorem 3.2 (Cochran's). Suppose $\mathbf{y}_{n \times 1} \sim N(\boldsymbol{\mu}, \mathbf{I})$, and $(\mathbf{A}_i)$ are symmetric matrices with $\sum_i \mathbf{A}_i = \mathbf{I}$. Let $n_i = \text{rank}(\mathbf{A}_i)$ and $\delta_i = \boldsymbol{\mu}' \mathbf{A}_i \boldsymbol{\mu}$. Then, $Q_i = \mathbf{y}_i' \mathbf{A}_i \mathbf{y}_i$ are independent $\chi_{n_i}^2(\delta_i)$ random variables $\iff \sum_i n_i = n$. □

Example 3.4. Suppose $\mathbf{y}_{n \times 1} \sim N(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$, and define $\mathbf{x}_{n \times 1} = \sigma^{-1} \mathbf{y} \sim N(\sigma^{-1} \boldsymbol{\mu}, \mathbf{I})$. Let $\mathbf{P}$ denote projection onto a linear subspace V of dimension m , and $\mathbf{Q} = \mathbf{I} - \mathbf{P}$ projection onto the orthogonal complement $V^\perp$. Now since $n_1 = m$ and $n_2 = n - m$, with $n_1 + n_2 = n$, by Cochran's theorem $\mathbf{y}' \mathbf{P} \mathbf{y} \sim \chi_m^2(\delta_1)$ and $\mathbf{y}' \mathbf{Q} \mathbf{y} \sim \chi_{n-m}^2(\delta_2)$ are independent, with $\delta_1 = \sigma^{-2} \cdot \boldsymbol{\mu}' \mathbf{P} \boldsymbol{\mu}$ and $\delta_2 = \sigma^{-2} \cdot \boldsymbol{\mu}' \mathbf{Q} \boldsymbol{\mu}$. ♠

Example 3.5. Suppose $\mathbf{y} \sim N(\mathbf{0}, \sigma^2 \mathbf{I})$. Let $\mathbf{P}_1$ and $\mathbf{P}_2$ denote projections onto orthogonal spaces $\mathbf{P}_1 \mathbf{P}_2 = \mathbf{0}$. Define:

$$\mathbf{z}_1 = \sigma^{-1} \mathbf{P}_1 \mathbf{y} \sim N(\mathbf{0}, \mathbf{P}_1), \quad \mathbf{z}_2 = \sigma^{-1} \mathbf{P}_2 \mathbf{y} \sim N(\mathbf{0}, \mathbf{P}_2).$$

The joint distribution of $\mathbf{z}_1$ and $\mathbf{z}_2$ is expressible as:

$$\begin{pmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 \end{pmatrix} = \sigma^{-1} \begin{pmatrix} \mathbf{P}_1 \\ \mathbf{P}_2 \end{pmatrix} \mathbf{y} = \mathbf{P}^* \mathbf{y} \sim N \left\{ \begin{pmatrix} \mathbf{0} \\ \mathbf{0} \end{pmatrix}, \begin{pmatrix} \mathbf{P}_1 & \mathbf{0} \\ \mathbf{0} & \mathbf{P}_2 \end{pmatrix} \right\}$$

Since $\mathbf{z}_1$ and $\mathbf{z}_2$ are independent jointly normal random variables, the quadratic forms:

$$\mathbf{z}_1' \mathbf{z}_1 = \sigma^{-2} \mathbf{y}' \mathbf{P}_1 \mathbf{y}, \quad \mathbf{z}_2' \mathbf{z}_2 = \sigma^{-2} \mathbf{y}' \mathbf{P}_2 \mathbf{y}$$

are independent χ^2 random variables. Moreover, the ratio:

$$\frac{\mathbf{z}_1' \mathbf{z}_1 / n_1}{\mathbf{z}_2' \mathbf{z}_2 / n_2} = \frac{\mathbf{y}' \mathbf{P}_1 \mathbf{y} / n_1}{\mathbf{y}' \mathbf{P}_2 \mathbf{y} / n_2} \sim F_{n_1, n_2}(0)$$

follows a central F distribution with n_1 and n_2 degrees of freedom.

